



NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Third Semester B.Tech (CBCS) Degree Examinations
 December 2024

ME1006-1 – MANUFACTURING PROCESSES
 (For EC)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS

20 Marks

1. **What is the purpose of the core in sand casting** _____
 A) To create the mold cavity
 B) To remove the casting from the mold
 C) To melt the metal
 D) To create internal features in the casting
2. **Why should the moulding sand be porous?**
 A) For gases to enter
 B) For gases to escape
 C) For water to enter
 D) For water to escape
3. **What is the function of a riser in a sand casting?**
 A) Provide good mobility to the molten metal
 B) To provide differential cooling rates in specific areas of the casting
 C) To develop holes or hollow cavities in the casting
 D) Prevent cavities due to shrinkage
4. **Runners, gates and risers are attached to the pattern to** _____
 A) Provide ease of cooling
 B) Provide feed to the molten material
 C) Provide design to the casting
 D) Enhance the finishing of the casting
5. **Which of the following statement is true?**
 A) It is possible to pour only ferrous metals for casting in Nobake process
 B) It is possible to pour only non-ferrous metals for casting in Nobake process
 C) Both ferrous as well as non-ferrous metals can be used for casting through Nobake process
 D) Neither ferrous, nor non-ferrous metals are used for casting in nobake process
6. **Which of the following compound forms the bond between sand particles in carbon dioxide moulding process?**
 A) Sodium
 B) Amorphous Silica
 C) Dehydrated water
 D) Carbon monoxide
7. **Which of the following methods of casting is the fastest method for the production of constant cross-section shapes of casting?**
 A) Continuous casting
 B) Centrifugal casting
 C) Permanent mould casting
 D) Die casting
8. **True centrifugal casting method is usually used to make** _____
 A) Bent pipes
 B) Nuts
 C) Bolts
 D) Hollow pipes
9. **What is the shape of the billet used in extrusion process?**
 A) Cubical
 B) Spherical
 C) Cylindrical
 D) Pyramidal
10. **Which of the following is also called as forward hot extrusion?**
 A) Direct extrusion
 B) Indirect extrusion
 C) Impact extrusion
 D) Hydrostatic extrusion

11. Which of the following operations involves reduction in length?
 - A) Upsetting
 - B) Drawing down
 - C) Setting down
 - D) Punching
12. Which of the following transforms compacted mechanical bonds between powder particles into metallurgical bonds?
 - A) SHP
 - B) Metal Injection Molding
 - C) Sintering
 - D) Mechanical alloying
13. In polymer extrusion process, breaker plate is used to
 - A) Melt the polymer
 - B) Pressurise the polymer
 - C) Heal the polymer
 - D) Filter the polymer melt
14. _____ process is used to make plastic bottles.
 - A) Blow molding
 - B) Rotomolding
 - C) Extrusion
 - D) Injection molding
15. Vacuum is used to shape the plastic in
 - A) Blow molding
 - B) Thermoforming
 - C) Extrusion
 - D) Injection molding
16. Which is the disadvantage of a ceramic material?
 - A) Hard
 - B) Brittle
 - C) Chemically stable
 - D) Refractory
17. What is the purpose of the flux coating on the electrode in shielded metal arc welding?
 - A) To create a stable arc
 - B) To provide filler metal for the weld
 - C) To shield the weld from atmospheric gases
 - D) To cool the weld after it is formed
18. Which metal is commonly used as a filler metal in soldering?
 - A) Copper
 - B) Aluminum
 - C) Tin
 - D) Zinc
19. What type of torch is commonly used for brazing?
 - A) MIG torch
 - B) TIG torch
 - C) Oxy-acetylene torch
 - D) Stick welding torch
20. What is the purpose of a horn in ultrasonic welding?
 - A) To hold the parts to be welded
 - B) To generate high-frequency sound waves
 - C) To provide heat for welding
 - D) To apply pressure for welding

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1.	a) What are the stages involved in any metal casting process? Describe.	8	L*2	1	1
	b) Describe the process of shell moulding with a neat schematic sketch.	8	L2	2	1
2.	a) With a schematic representation illustrate the unique features of, i) Match plate pattern ii) Gated pattern	8	L2	1	1
	b) Write a note on the following special casting techniques i) Evaporative pattern casting ii) Squeeze casting	8	L2	2	1
3.	a) With a pictorial representation, describe the conditions in which following core types are used. i) Ram up core ii) Hanging core	8	L2	1	1
	b) Distinguish between the features of true centrifugal casting and semi centrifugal casting.	8	L2	2	1
Unit – II					
4.	a) Compare the process of direct and indirect extrusion with a schematic sketch.	8	L3	3	1

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|-------|---|---|----|---|---|
| b) | What are polymers? Write the features and properties of thermoplastics and thermosets. | 8 | L2 | 4 | 1 |
| 5. a) | Write a note on following forming processes | | | | |
| | i) Rod drawing | 8 | L3 | 3 | 1 |
| | ii) Tube drawing | | | | |
| b) | Write a note on following ceramic processing methods. | | | | |
| | i) Pressing | 8 | L2 | 4 | 1 |
| | ii) Sintering | | | | |
| 6. a) | Narrate the functioning of compound die set up and explain how it is helpful in increasing the production rate. | 8 | L3 | 3 | 1 |
| b) | With a schematic representation explain the Blow moulding technique for plastic materials. | 8 | L2 | 4 | 1 |

Unit – III

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|-------|--|---|----|---|---|
| 7. a) | Give the detailed classification of Metal joining techniques. Write the principle of Metal Arc welding | 8 | L2 | 5 | 1 |
| b) | Write the nature of following types of flames obtained from gas welding torch, | | | | |
| | i) Carburizing flame | 8 | L2 | 5 | 1 |
| | ii) Oxidizing flame | | | | |
| 8. a) | Write the causes for following welding defects | | | | |
| | i) Porosity | | | | |
| | ii) Undercuts | | | | |
| | iii) Distortion | 8 | L2 | 5 | 1 |
| | iv) Inclusions | | | | |
| b) | With a schematic representation, explain the construction and working of a Ultrasonic welding process. | 8 | L2 | 5 | 1 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
